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Test 2016, answers

Database Management Systems (The University of the South Pacific)

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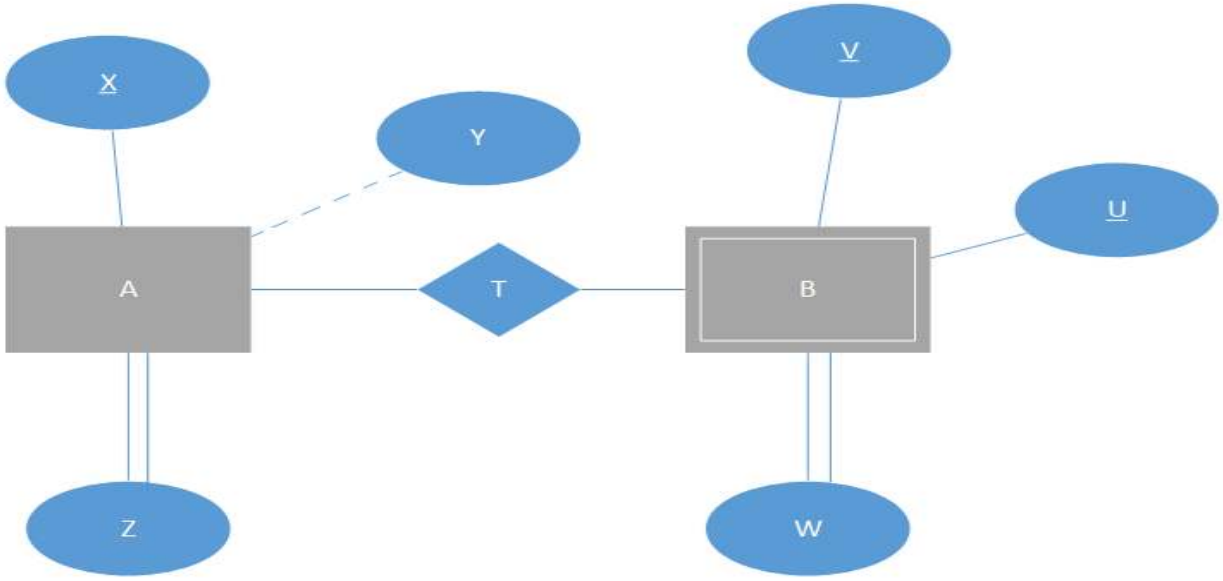
IS222 S1 2016 - MIDTERM TEST

Note: See your test paper for details on the questions. Not all questions will be listed here in full.

Section B (21 Marks)

1) Consider the figure given below and answer the questions that follow. **(1.5mks total)**



What type of entity/attributes are depicted for the following in the figure above? Only **one word answers** are required in the spaces given.

Solution ***(0.5mk each)***

- B is an entity while the others are attributes. Only one word answers suffice. However, if B is called a weak attribute for instance, then it would be incorrect. The same applies for Y an Z which should not be referred to as entities.

i) **B** - _____weak_____

ii) **Y** - _____derived_____

iii) **Z** - _____multivalued_____

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2) Fill in the blanks in the statement below in the spaces given. **(1mk total)**

OLTP databases are based largely on _____(i)_____ operations whilst OLAP databases are based mostly on _____(ii)_____ operations.

Solution ***(0.5mk each)***

i) _____write_____

ii) _____read/select_____

3) Consider the tables given below……. **(0.5 mk)**

$$\left(\prod_{\text{name, age}} (\text{Manager}) - \prod_{\text{name, age}} (\text{Customer}) \right) \times \text{Project}$$

Solution

Name	Age	P_ID
Robinson	37	X
Robinson	37	Y

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4) Read the scenario given below and answer the questions that follow…….

Scenario:

You have been hired to design a small database for a new company called StarkCorp. StarkCorp only need a few details tracked in their database which deals with its customers, particular products and invoices for the moment……

i) List the business rules that you have used to design your database. **[4mks total]**

- **2mks per rule** *(0/1mk each side of the rule)*
  - *Cardinalities must be correct*
- Note that synonyms are acceptable for cardinalities eg. one and only one, and exactly one are treated as synonyms. Use discretion where necessary
- Redundant rules (i.e. those that are irrelevant or repeated) to be penalized (add up to -1mk max to the total). Use discretion
- **Lowest score possible is 0mk for this question overall**

Rules

1)	**Each customer** has been issued **one or more invoices**. **Each invoice** is issued to **exactly one customer**.
2)	**Each invoice** consists of **only one product.** **Each product** generates **zero or more invoices.**

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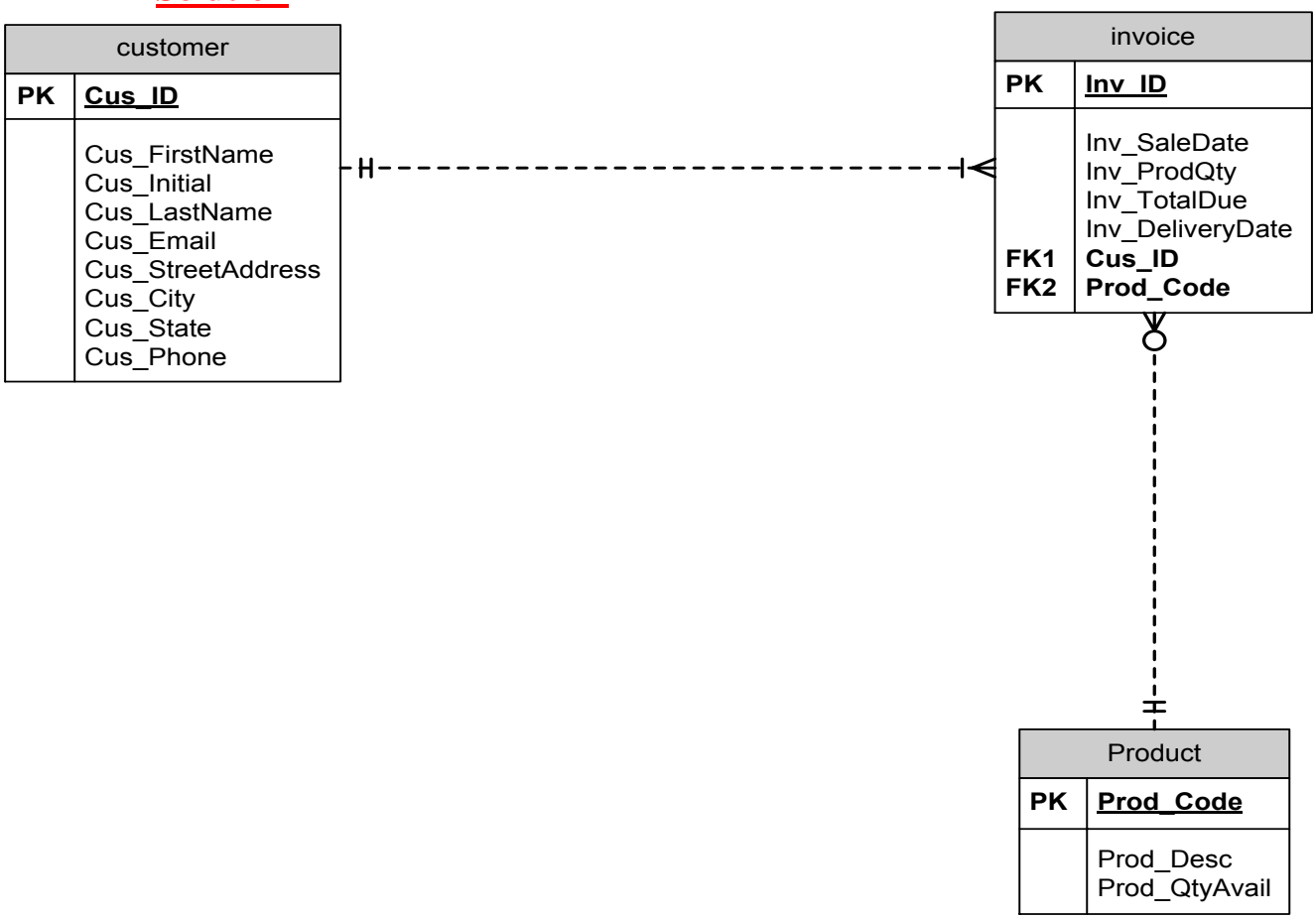
ii) Draw your Crow's Foot Entity Relationship Diagram (or ERD)…….

Solution



ERD (14mks total)

For each entity - ***(0.5mk altogether*** *for the correct PK and FK, with the* ***rest of the particular entity's mark total below*** *for its remaining attributes apart from the PK/FK attributes)*

- **4.5 mks invoice**
- **4.5 mks product**
- **3 mks customer**

2mks relationships (total)

- ***1mk each relationship*** *(0.5mk strength and 0.5mk for the relationship if everything is correct)*
- Naming conventions for attributes not followed (-0.5mk from total)
- Entities not named (-1mk from total)

- **Lowest score possible for ERD is 0mk**

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Section C (11 Marks)

1. **[3 mks]**

   (1mk each)

	Actual Relation	**Functional Dependencies**	**Decomposition of Z**	**Lossless decomposition? (yes/no)**
(i)	Z (P, Q, R)	P → Q Q → R	R1 (P, Q) R2 (Q, R)	**Yes**
(ii)	Z (P, Q, R)	R → Q P → Q	R1 = (P, Q) R2 = (P, R)	**Yes**
(iii)	Z(P,Q,R,S,T, U)	PQ→RS S→P	R1(P,Q,R,U) R2(P,Q,S,T)	**No**

2. **[2mks]**

   (0.5mk each)

Functional Dependency	**Trivial? (yes/no)**
O → P	**No**
OP → P	**Yes**
PQR → ST	**No**
OP → O	**Yes**

3. Consider the following relational schema, Z, and its set of functional dependencies, F……..

   i. List the candidate key(s) for Z……. **(2mks total)**

   <u>Solution</u> ***(0.5mk each)***

   QSPR, QSPU, QSRT, QSTU

   There are 4 keys. Students would number each.

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ii. Initiating the Lossless or Nonadditive BCNF decomposition of Z…… **(3mks total)**

Marker Note

- **1mk per table** *(0.5mk for all correct attributes without redundant ones and 0.5mk for dependency)*
    - **If attributes in a table are wrong, give 0 for that table**

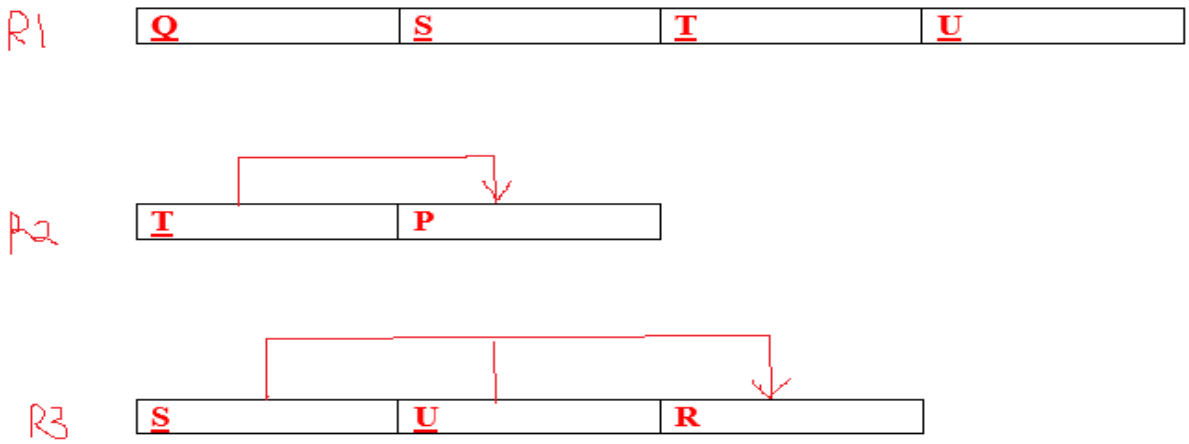
- Must show the key for R1 for full credit for R1

- Reduce marks for redundant tables from the total if there are more than 3 tables (-1mk per table).

- Tables can be named anything reasonable, doesn't need the exact names as given here. If table names are missing (-0.5mk from the total)

- **Lowest score possible is 0 overall**

Solution



iii. Following the decomposition in part (ii) above, which candidate key(s) of the original relation, Z, would be expected to be fully preserved or still remain in the decomposition?

Solution QSTU ***(0/1 mk)***

Only one remains preserved

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